Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Ovary

Date: 22 May 2025
Time: 12:24:09 PM

Female

Gene Symbols	Entrez Gene IDs	Probe IDs	BMD_{rd25} ($BMDL_{rd25}$ - $BMDU_{rd25}$) in ppm	Maximum Fold Change	Direction of Expression Change
Fosl1	25445	FOSL1_8659	127.2 (22.7 - 291.4)	11.4	DOWN
Apoa1	25081	APOA1_33150	187.9 (47.2 - 378.4)	3.2	DOWN
S100a9	94195	S100A9_9775	199.6 (78.7 - 432.2)	1.8	UP
Acss1	296259	ACSS1_32386	213.3 (63.5 - 449.8)	1.8	UP
Cxcl12	24772	CXCL12_8410,CXCL12_32815	246.7 (103.1 - 433.5)	1.7	UP
Tnfrsf12a	302965	TNFRSF12A_10049	255.3 (113.7 - 412.3)	3.9	DOWN
Slc26a2	117267	SLC26A2_33072	258.7 (84.0 - 430.3)	1.7	UP
Aen	361594	AEN_7998	291.1 (93.4 - 2345.0)	1.5	DOWN
Vnn1	29142	VNN1_10157	310.0 (166.8 - 452.8)	6.9	UP
Slc51a	303879	SLC51A_33157	314.8 (131.0 - 483.1)	3.1	DOWN

rd25 = 25% relative deviation

Annotations:

Descriptions of orthologous human genes are shown due to the increased detail that is available in public resources such as UniprotKB (http://www.uniprot.org/uniprot/) and Entrez Gene (https://www.ncbi.nlm.nih.gov/gene/). Human UniprotKB was used as the primary resource due to the greater breadth of annotation and depth of functional detail that is provided. Rat UniprotKB was used as the second resource if the primary source did not provide a detailed description of function. Human Entrez gene summary was used as third resource. Rat Entrez gene summary was used as the fourth resource.
Gene definition version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

Fosl1: HumanUniprotfunction(HumanFOSL1): The Fos gene family consists of 4 members: FOS, FOSB, FOSL1, and FOSL2. These genes encode leucine zipper proteins that can dimerize with proteins of the JUN family, thereby forming the transcription factor complex AP-1. As such, the FOS proteins have been implicated as regulators of cell proliferation, differentiation, and transformation. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Jul 2014]

Apoa1: Human Uniprot function (Human APOA1): Participates in the reverse transport of cholesterol from tissues to the liver for excretion by promoting cholesterol efflux from tissues and by acting as a cofactor for the lecithin cholesterol acyltransferase (LCAT). As part of the SPAP complex, activates spermatozoa motility. \{ECO:0000269;PubMed:1909888\}.

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S100a9: Human Uniprot function (Human S100A9): S100A9 is a calcium- and zinc-binding protein which plays a prominent role in the regulation of inflammatory processes and immune response. It can induce neutrophil chemotaxis, adhesion, can increase the bactericidal activity of neutrophils by promoting phagocytosis via activation of SYK, PI3K/AKT, and ERK1/2 and can induce degranulation of neutrophils by a MAPK-dependent mechanism. Predominantly found as calprotectin (S100A8/A9) which has a wide plethora of intra- and extracellular functions. The intracellular functions include: facilitating leukocyte arachidonic acid trafficking and metabolism, modulation of the tubulin-dependent cytoskeleton during migration of phagocytes and activation of the neutrophilic NADPH-oxidase. Activates NADPH-oxidase by facilitating the enzyme complex assembly at the cell membrane, transferring arachidonic acid, an essential cofactor, to the enzyme complex and S100A8 contributes to the enzyme assembly by directly binding to NCF2/P67PHOX. The extracellular functions involve proinflammatory, antimicrobial, oxidant-scavenging and apoptosis-inducing activities. Its proinflammatory activity includes recruitment of leukocytes, promotion of cytokine and chemokine production, and regulation of leukocyte adhesion and migration. Acts as an alarmin or a danger associated molecular pattern (DAMP) molecule and stimulates innate immune cells via binding to pattern recognition receptors such as Toll-like receptor 4 (TLR4) and receptor for advanced glycation endproducts (AGER). Binding to TLR4 and AGER activates the MAP-kinase and NF-kappa-B signaling pathways resulting in the amplification of the proinflammatory cascade. Has antimicrobial activity towards bacteria and fungi and exerts its antimicrobial activity probably via chelation of Zn(2+) which is essential for microbial growth. Can induce cell death via autophagy and apoptosis and this occurs through the cross-talk of mitochondria and lysosomes via reactive oxygen species (ROS) and the process involves BNIP3. Can regulate neutrophil number and apoptosis by an anti-apoptotic effect; regulates cell survival via ITGAM/ITGB and TLR4 and a signaling mechanism involving MEK-ERK. Its role as an oxidant scavenger has a protective role in preventing exaggerated tissue damage by scavenging oxidants. Can act as a potent amplifier of inflammation in autoimmunity as well as in cancer development and tumor spread. Has transnitrosylase activity; in oxidatively-modified low-densitity lipoprotein (LDL(ox))-induced S-nitrosylation of GAPDH on 'Cys-247' proposed to transfer the NO moiety from NOS2/iNOS to GAPDH via its own S-nitrosylated Cys-3. The iNOS-S100A8/A9 transnitrosylase complex is proposed to also direct selective inflammatory stimulus-dependent S-nitrosylation of multiple targets such as ANXA5, EZR, MSN and VIM by recognizing a [IL]-x-C-x-x-[DE] motif. \{ECO:0000269;PubMed:12626582,ECO:0000269;PubMed:15331440, ECO:0000269;PubMed:15598812, ECO:0000269;PubMed:15642721, ECO:0000269;PubMed:16258195,ECO:0000269;PubMed:19087201, ECO:0000269;PubMed:19122197, ECO:0000269;PubMed:19402754, ECO:0000269;PubMed:19534726, ECO:0000269;PubMed:19935772, ECO:0000269;PubMed:20103766, ECO:0000269; PubMed:21325622, ECO:0000269;PubMed:21487906, ECO:0000269;PubMed:22363402, ECO:0000269;PubMed:22804476, ECO:0000269;PubMed:22808130, ECO:0000269;PubMed:25417112, ECO:0000269;PubMed:8423249\}

Acss1: Human Uniprot function (Human ACSS1): Catalyzes the synthesis of acetyl-CoA from short-chain fatty acids (PubMed:16788062). Acetate is the preferred substrate (PubMed:16788062). Can also utilize propionate with a much lower affinity (By similarity). Provides acetyl-CoA that is utilized mainly for oxidation under ketogenic conditions (By similarity). Involved in thermogenesis under ketogenic conditions, using acetate as a vital fuel when carbohydrate availability is insufficient (By similarity). \{ECO:0000250;UniProtKB:Q99NB1, ECO:0000269;PubMed:16788062\}.

Cxcl12: Human Uniprot function (Human CXCL12): Chemoattractant active on T-lymphocytes and monocytes but not neutrophils. Activates the C-X-C chemokine receptor CXCR4 to induce a rapid and transient rise in the level of intracellular calcium ions and chemotaxis. SDF-1-beta(3-72) and SDF-1-alpha(3-67) show a reduced chemotactic activity. Binding to cell surface proteoglycans seems to inhibit formation of SDF-1-alpha(3-67) and thus to preserve activity on local sites. Also binds to atypical chemokine receptor ACKR3, which activates the beta-arrestin pathway and acts as a scavenger receptor for SDF-1. Binds to the allosteric site (site 2) of integrins and activates integrins ITGAV:ITGB3, ITGA4:ITGB1 and ITGA5:ITGB1 in a CXCR4-independent manner (PubMed:29301984). Acts as a positive regulator of monocyte migration and a negative regulator of monocyte adhesion via the LYN kinase. Stimulates migration of monocytes and T-lymphocytes through its receptors, CXCR4 and ACKR3, and decreases monocyte adherence to surfaces coated with ICAM-1, a ligand for beta-2 integrins. SDF1A/CXCR4 signaling axis inhibits beta-2 integrin LFA-1 mediated adhesion of monocytes to ICAM-1 through LYN kinase. Inhibits CXCR4-mediated infection by T-cell line-adapted HIV-1. Plays a protective role after myocardial infarction. Induces down-regulation and internalization of ACKR3 expressed in various cells. Has several critical functions during embryonic development; required for B-cell lymphopoiesis, myelopoiesis in bone marrow and heart ventricular septum formation. Stimulates the proliferation of bone marrow-derived B-cell progenitors in the presence of IL7 as well as growth of stromal cell-dependent pre-B-cells (By similarity). \{ECO:0000250;UniProtKB:P40224, ECO:0000269;PubMed:11069075, ECO:0000269;PubMed:11069075, ECO:0000269;PubMed:11859124, ECO:0000269;PubMed:16107333, ECO:0000269;PubMed:18802065, ECO:0000269;PubMed:19255243, ECO:0000269;PubMed:29301984, ECO:0000269;PubMed:8752281\}.

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Tnfrsf12a: Human Uniprot function (Human TNFRSF12A): Receptor for TNFSF12/TWEAK. Weak inducer of apoptosis in some cell types. Promotes angiogenesis and the proliferation of endothelial cells. May modulate cellular adhesion to matrix proteins. \{ECO:0000269;PubMed:11728344\}.

Slc26a2: Human Uniprot function (Human SLC26A2): Sulfate transporter. May play a role in endochondral bone formation. \{ECO:0000269;PubMed:7923357\}.

Aen: Human Uniprot function (Human AEN): Exonuclease with activity against single- and double-stranded DNA and RNA. Mediates p53-induced apoptosis. When induced by p53 following DNA damage, digests double-stranded DNA to form single-stranded DNA and amplifies DNA damage signals, leading to enhancement of apoptosis. \{ECO:0000269;PubMed:16171785, ECO:0000269;PubMed:18264133\}.

Vnn1: Human Uniprot function (Human VNN1): Amidohydrolase that hydrolyzes specifically one of the carboamide linkages in D-pantetheine thus recycling pantothenic acid (vitamin B5) and releasing cysteamine. \{ECO:0000269;PubMed:10567687, ECO:0000269;PubMed:11491533, ECO:0000269;PubMed:25478849\}

Slc51a: Human Uniprot function (Human SLC51A): Essential component of the Ost-alpha/Ost-beta complex, a heterodimer that acts as the intestinal basolateral transporter responsible for bile acid export from enterocytes into portal blood. Efficiently transports the major species of bile acids. \{ECO:0000269;PubMed:16317684\}.

** END OF REPORT **